ITBDA – 14
Documentation

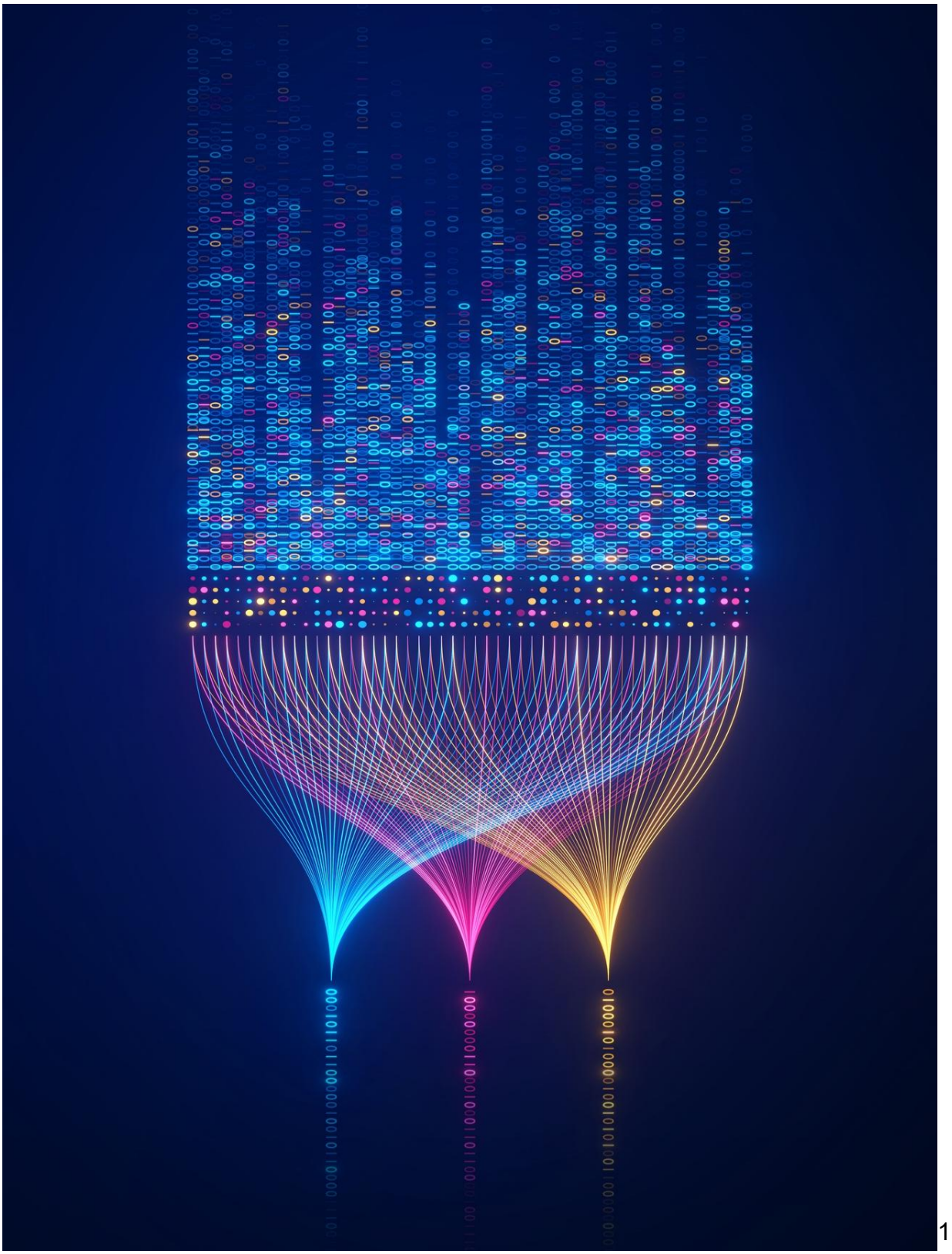


Table of Contents

Question 1 3
Question 2 5
Question 3 9
Question 4 11
Bibliography 12

Question 1

1.1)

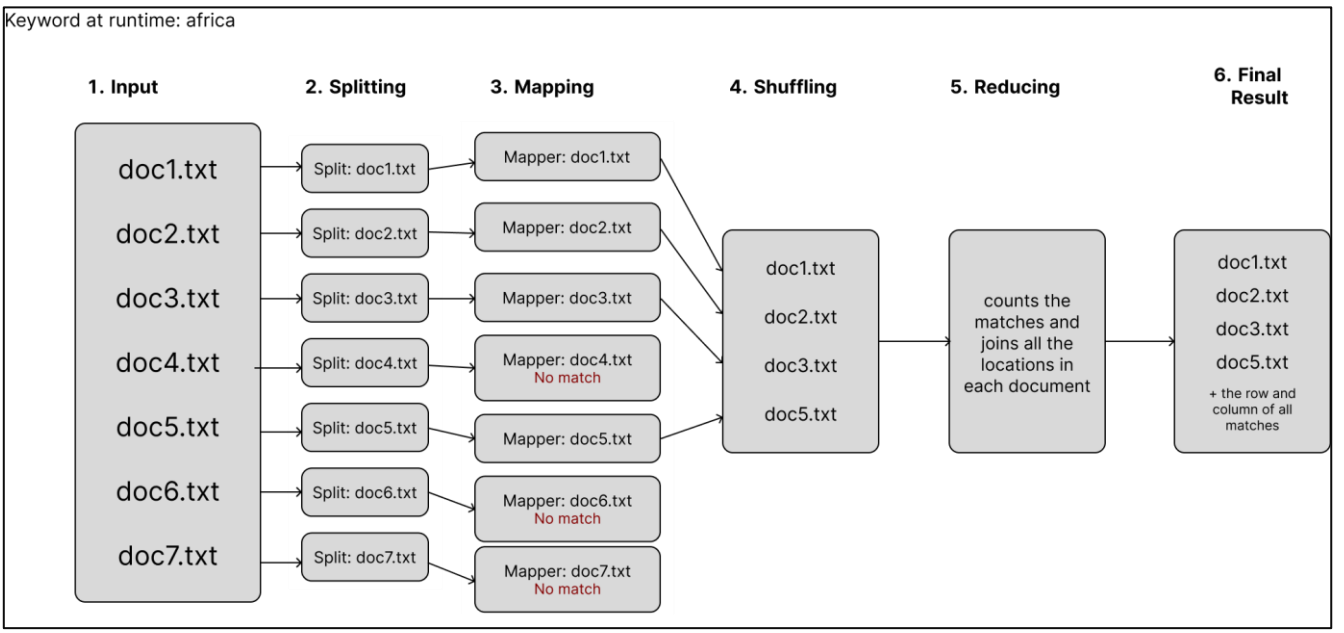


Figure 1: Logical layout of the MapReduce keyword search job.

The figure above shows how the keyword search will be carried out as a MapReduce job. The job moves through four main stages: splitting, mapping, shuffling, and reducing (Traininghub.IO, 2023).

Input and Splitting: The seven articles are placed in an input folder, and Hadoop will divide the input into splits, with one map task started for each split (Apache Software Foundation, 2023). Because each article is smaller than the 128 MB HDFS block (Apache Software Foundation, 2026), each article will become its own split.

Keyword Input: The keyword is not hardcoded in this situation. It will be entered when the job is run and passed to every mapper as an environment variable using Hadoop Streaming, which will allow the mapper and reducer to be written as R scripts.

Mapping: Each mapper will read its article line by line, using a line counter for the row and the start position of each match for the column. The line and the keyword will be converted to lower case so the search is case-insensitive, and every match will be emitted as a key-value pair. Any articles with no match will emit nothing.

Shuffling: Hadoop will collect the output of all the mappers and sort it by key, so all the locations for the same article are grouped together before they reach the reducer (Apache Software Foundation, 2023; GeeksforGeeks, 2025).

Reducing and Final Result: The reducer will count the matches for each article and join their locations into one output record. The keyword “africa” gives doc1.txt, doc2.txt, doc3.txt, and doc5.txt, each with the relevant row and column of each match.

1.2) Please see the attached R script (Question 1).

1.3) Please see the attached R script (Question 1).

1.4) Please see the attached R script (Question 1).

```
2026-11-08 15:38:50,809 INFO streaming.StreamJob: Output directory: output
Keyword search results for 'africa':
 document matches                                                                locations
 doc1.txt       3                                                    (1,64) (3,11) (3,95)
 doc2.txt       5                                  (1,57) (3,32) (33,24) (58,5) (58,81)
 doc3.txt       9 (1,38) (3,7) (5,7) (40,146) (46,7) (48,112) (50,142) (52,86) (54,44)
 doc5.txt       5                                (1,62) (3,88) (23,121) (38,27) (48,217)

C:\q1_mapreduce>
```

Figure 2: Keyword search output for "africa"

Note: The data exploration is in explore_corpus.R. To run the MapReduce job, run keyword_search.R with a keyword (e.g. Rscript keyword_search.R africa). It runs mapper.R and reducer.R through Hadoop, so those two don’t need to be run separately.

Question 2

2.1) Please see the attached R script (Question 2).

2.2) Please see the attached R script (Question 2).

2.3) Please see the attached R script (Question 2).

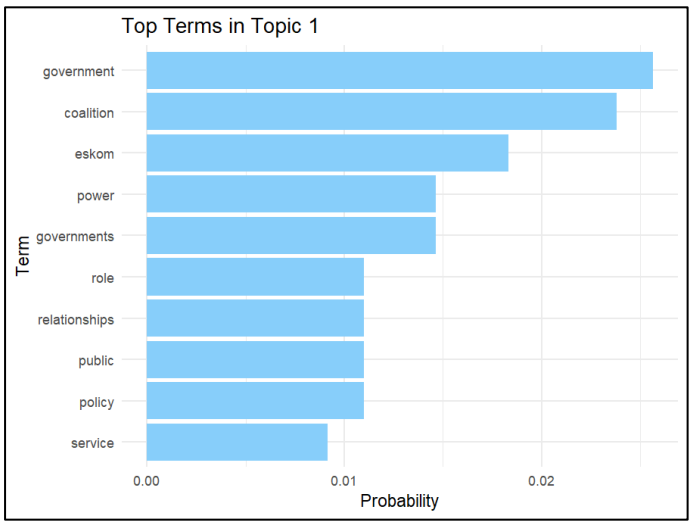


Figure 3: Bar Plot for Topic 1

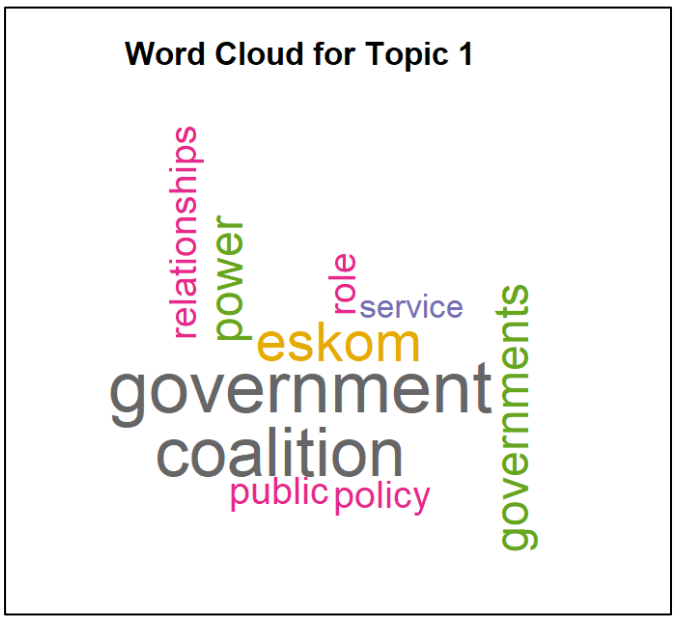


Figure 4: Word Cloud for Topic 1

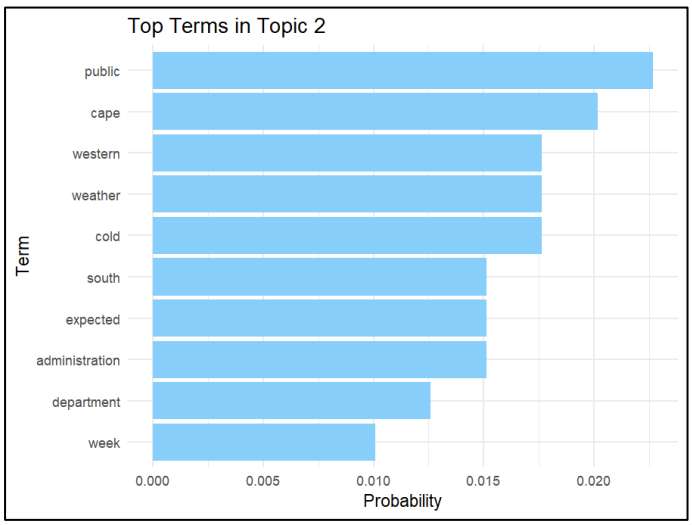


Figure 5: Bar Plot for Topic 2

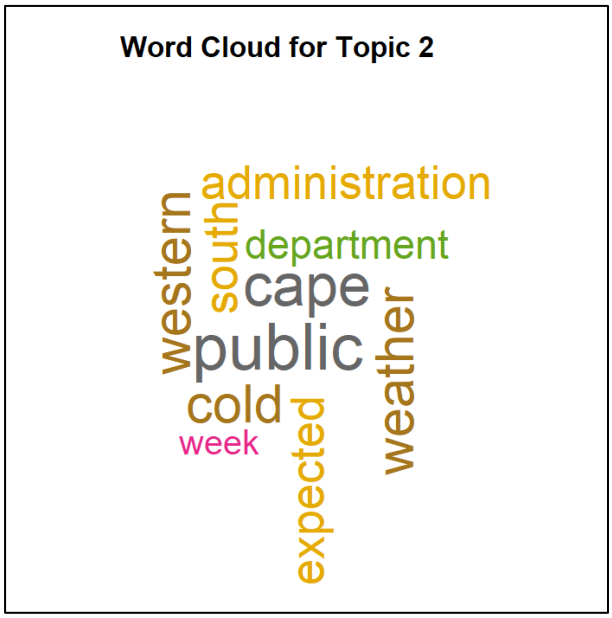


Figure 6: Word Cloud for Topic 2

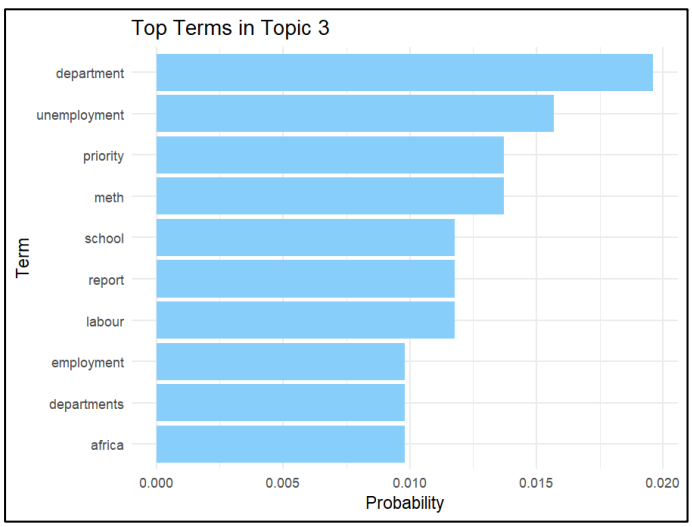


Figure 7: Bar Plot for Topic 3

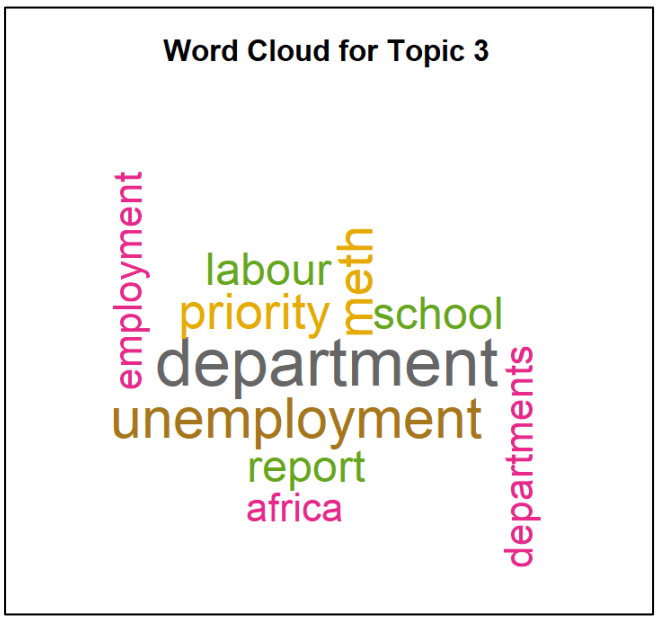


Figure 8: Word Cloud for Topic 3

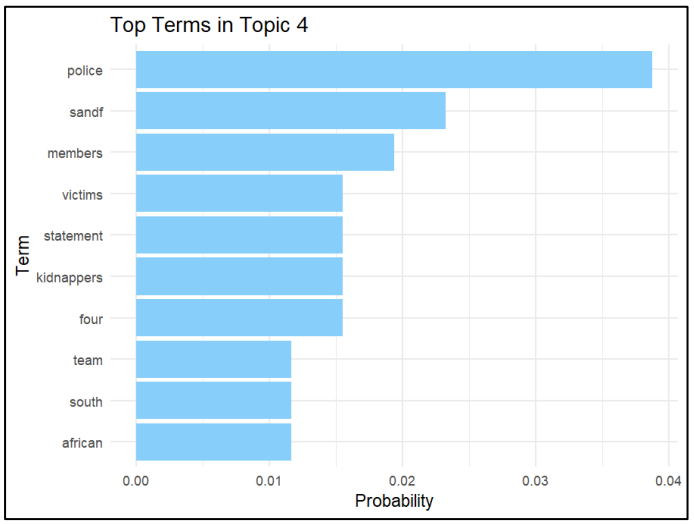


Figure 9: Bar Plot for Topic 4

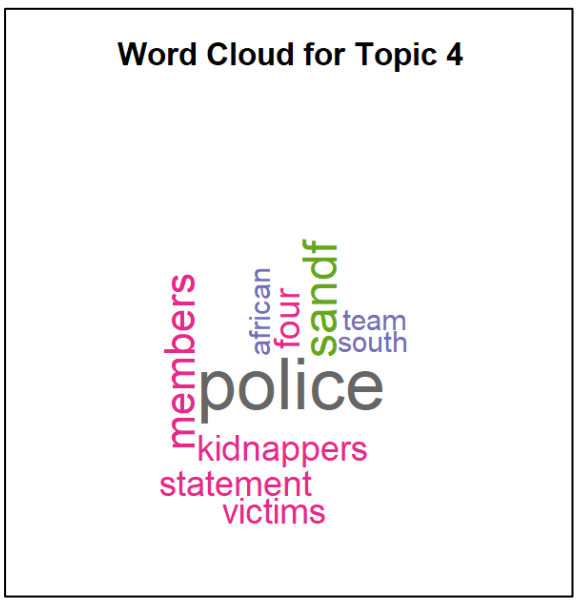


Figure 10: Word Cloud for Topic 4

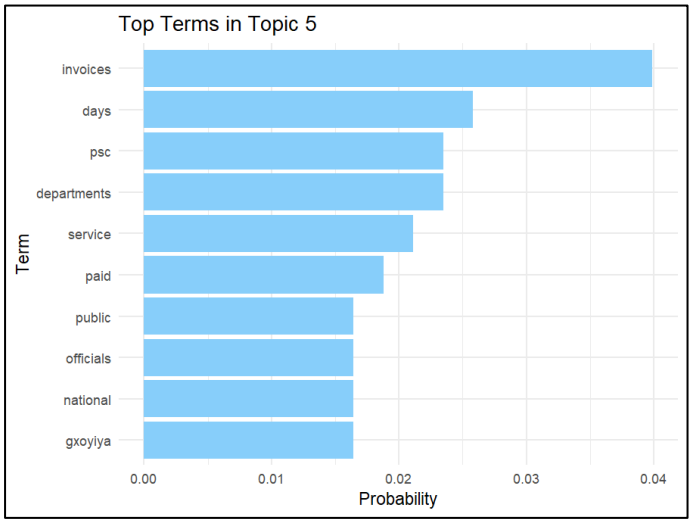


Figure 11: Bar Plot for Topic 5

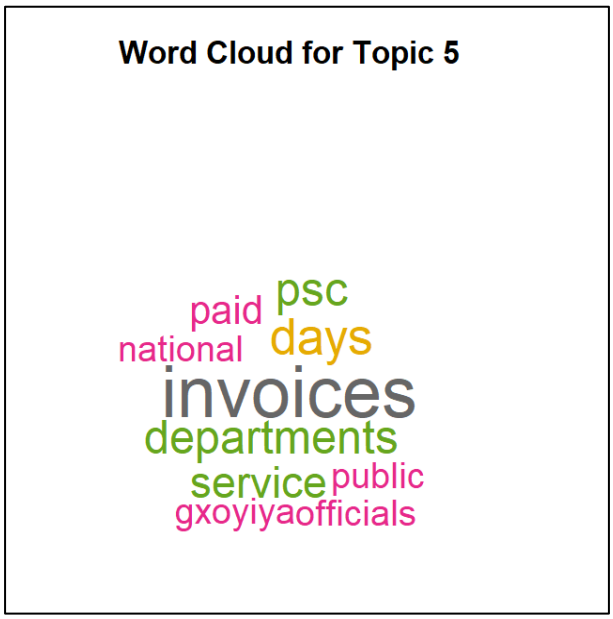


Figure 12: Word Cloud for Topic 5

Question 3

3.1) Please see the attached R script (Question 3).

3.2) Please see the attached R script (Question 3).

3.3) Two techniques were used to summarise the 20 sentences of the page description: word frequency scoring and TF-IDF scoring. The top three sentences from each technique form the summary. The R code is in the attached R file, Question3.R, and in the GitHub repository.

Word frequency summary (sentences 3, 4, and 10), 7.7% of the original words:

"The Motor Vehicle Collisions data tables contain information from all police reported motor vehicle collisions in NYC. View all Motor Vehicle Collisions data. Police officers complete form MV-104AN for all vehicle collisions."

TF-IDF summary (sentences 2, 5, and 7), 13.3% of the original words:

"Each row represents a crash event. The police report (MV104-AN) is required to be filled out for collisions where someone is injured or killed, or where there is at least $1000 worth of damage (https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/ny_overlay_mv-104an_rev05_2004.pdf). Due to success of the CompStat program, NYPD began to ask how to apply the CompStat principles to other problems."

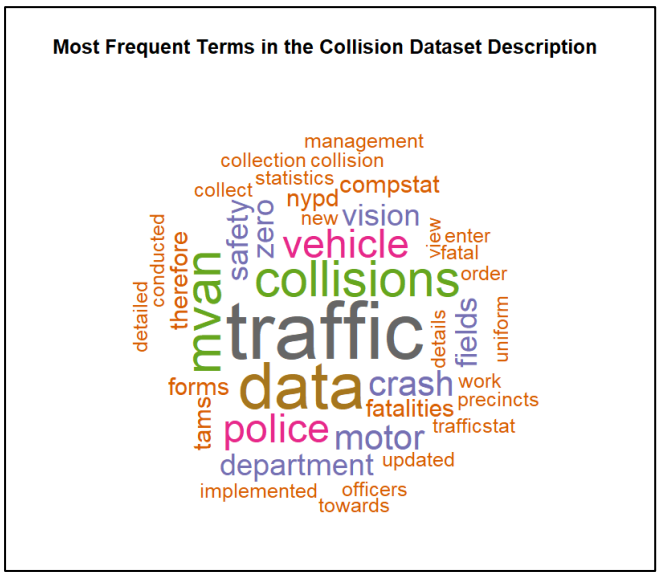


Figure 13: Word Cloud

The word cloud (Figure 13) shows that the description is about traffic collisions. "Traffic" is the largest word, followed by "data", "collisions", "vehicle", "crash", "police", and "motor". "Mvan" is the police form code MV-104AN, whose number and hyphen were removed during cleaning.

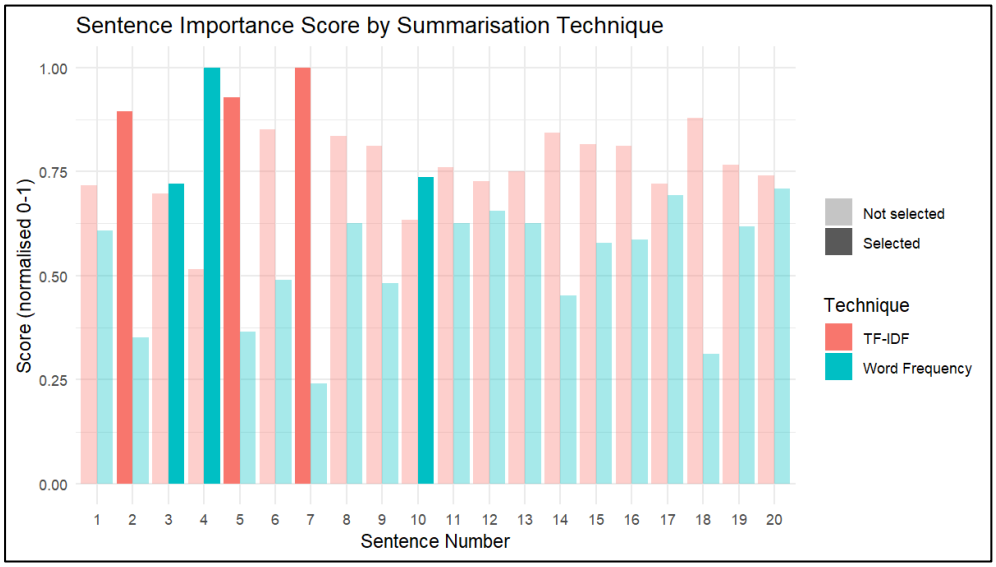


Figure 14: Sentence Scores by Method

The two techniques chose completely different sentences. Word frequency chose short sentences made of common terms (motor, vehicle, collisions, police). TF-IDF chose longer sentences with more specific details, such as crash events and the $1000 damage rule. Word frequency scores range from 0.25 to 1.0, while TF-IDF scores are mostly between 0.5 and 1.0.

Question 4

4.1) Please see the attached R script (Question 4).

4.2) Please see the attached R script (Question 4).

4.3) Please see the attached R script (Question 4).

Proof of database creation:



Figure 15: books_db in pgAdmin 4

	Title text	Price double precision	Rating integer	Availability text
1	A Light in the Attic	51.77	3	In stock
2	Tipping the Velvet	53.74	1	In stock
3	Soumission	50.1	1	In stock
4	Sharp Objects	47.82	4	In stock
5	Sapiens: A Brief History of Humankind	54.23	5	In stock
6	The Requiem Red	22.65	1	In stock
7	The Dirty Little Secrets of Getting Your Dream Job	33.34	4	In stock
8	The Coming Woman: A Novel Based on the Life of the Infamous Feminist, Victoria Woodhull	17.93	3	In stock
9	The Boys in the Boat: Nine Americans and Their Epic Quest for Gold at the 1936 Berlin Olym...	22.6	4	In stock
10	The Black Maria	52.15	1	In stock
11	Starving Hearts (Triangular Trade Trilogy, #1)	13.99	2	In stock
12	Shakespeare's Sonnets	20.66	4	In stock
13	Set Me Free	17.46	5	In stock

Figure 16: books_db table in pgAdmin 4

Interpretation of Queries:

The price of a book and its rating in this sample seem to be unrelated. The average price fluctuates across the star ratings. 3-star books have the highest average price (42.32) and 4-star books the lowest (31.11), while 1-star and 5-star books cost almost the same on average (40.02 and 39.75). The five most expensive books, priced between 52.15 and 57.25, are rated 3, 5, 1, 5, and 1, so the highest-priced books are not the best rated either. With only 20 books, and 3 to 6 in each rating group, one book can shift an average, so these results describe only the given sample.

4.4) Please see GitHub Repository link:

https://github.com/Jess871/ITBDA4---Project-2

Bibliography

Apache Software Foundation. 2022. *Hadoop Streaming* (version 3.3.4). Apache Hadoop, 29 July 2022. [online] Available at: https://hadoop.apache.org/docs/r3.3.4/hadoop-streaming/HadoopStreaming.html [Accessed: 17 October 2026].

Apache Software Foundation. 2023. *Hadoop: setting up a single node cluster* (version 3.3.5). Apache Hadoop, 15 March 2023. [online] Available at: https://hadoop.apache.org/docs/r3.3.5/hadoop-project-dist/hadoop-common/SingleCluster.html [Accessed: 17 October 2026].

Apache Software Foundation. 2023. *mapred-default.xml* (version 3.3.6). Apache Hadoop. [online] Available at: https://hadoop.apache.org/docs/r3.3.6/hadoop-mapreduce-client/hadoop-mapreduce-client-core/mapred-default.xml [Accessed: 18 October 2026].

Apache Software Foundation. 2023. *MapReduce tutorial* (version 3.3.6). Apache Hadoop, 18 June 2023. [online] Available at: https://hadoop.apache.org/docs/r3.3.6/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html [Accessed: 17 October 2026].

Apache Software Foundation. 2026. *HDFS architecture* (version 3.5.0). Apache Hadoop, 24 March 2026. [online] Available at: https://hadoop.apache.org/docs/current/hadoop-project-dist/hadoop-hdfs/HdfsDesign.html [Accessed: 17 October 2026].

Bobbitt, Z. 2021. How to Use the gsub() Function in R (With Examples). *Statology*, 24 June 2021. [online] Available at: https://www.statology.org/gsub-r/ [Accessed: 5 November 2026].

Bobbitt, Z. 2022. How to Replace NA with Median in R. *Statology*, 01 August 2022. [online] Available at: https://www.statology.org/r-replace-na-with-median/ [Accessed: 5 November 2026].

Bobbitt, Z. 2023. How to Perform Data Cleaning in R (With Example). *Statology*, 26 January 2023. [online] Available at: https://www.statology.org/data-cleaning-in-r/ [Accessed: 5 November 2026].

GeeksforGeeks. 2025. An Easy Approach to TF-IDF Using R. GeeksforGeeks, 23 July 2025. [online] Available at: https://www.geeksforgeeks.org/nlp/an-easy-approach-to-tf-idf-using-r/ [Accessed: 31 October 2026].

GeeksforGeeks. 2025. Convert R objects to/from JSON in jsonlite. *GeeksforGeeks*, 23 July 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/convert-r-objects-to-from-json-in-jsonlite/ [Accessed: 30 October 2026].

GeeksforGeeks. 2025. Generating Word Cloud in R Programming. *GeeksforGeeks*, 11 December 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/generating-word-cloud-in-r-programming/ [Accessed: 25 October 2026].

GeeksforGeeks. 2025. Map Reduce in Hadoop. *GeeksforGeeks*, 1 November 2025. [online] Available at: https://www.geeksforgeeks.org/big-data/map-reduce-in-hadoop/ [Accessed: 17 October 2026].

GeeksforGeeks. 2025. R - Charts and Graphs. *GeeksforGeeks*, 28 April 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/r-charts-and-graphs/ [Accessed: 25 October 2026].

GeeksforGeeks. 2025. Saving Graphs as Files in R. *GeeksforGeeks*, 23 July 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/saving-graphs-as-files-in-r/ [Accessed: 25 October 2026].

GeeksforGeeks. 2025. Splitting Strings in R programming - strsplit() method. *GeeksforGeeks*, 12 July 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/splitting-strings-in-r-programming-strsplit-method/ [Accessed: 31 October 2026].

GeeksforGeeks. 2025. Stop Word Removal In R. *GeeksforGeeks*, 23 July 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/stop-word-removal-in-r/ [Accessed: 31 October 2026]

GeeksforGeeks. 2025. Text Mining in R with tidytext. *GeeksforGeeks*, 23 July 2025. [online] Available at: https://www.geeksforgeeks.org/nlp/text-mining-in-r-with-tidytext/ [Accessed: 31 October 2026].

GeeksforGeeks. 2025. Text Summarization Techniques. *GeeksforGeeks*, 23 July 2025. [online] Available at: https://www.geeksforgeeks.org/nlp/text-summarization-techniques/ [Accessed: 31 October 2026].

GeeksforGeeks. 2025. Web Scraping using R Language. *GeeksforGeeks*, 06 August 2025. [online] Available at: https://www.geeksforgeeks.org/r-language/web-scraping-using-r-language/ [Accessed: 30 October 2026].

GeeksforGeeks. 2026. Create Database in PostgreSQL. *GeeksforGeeks*, 17 September 2026. [online] Available at: https://www.geeksforgeeks.org/postgresql/create-database-postgresql/ [Accessed: 6 November 2026].

JSON-LD.org. n.d. JSON for Linking Data. *json-ld.org*, n.d. [online] Available at: https://json-ld.org/ [Accessed: 30 October 2026].

Khushalani, A.S. 2020. Topic Modeling using R. *Medium*, 23 February 2020. [online] Available at: https://medium.com/@AnjaliKhushalani/topic-modeling-using-r-b90d895585b6 [Accessed: 24 October 2026].

Kingl, A. 2018. Web Scraping in R: rvest Tutorial. *DataCamp*, 27 February 2018. [online] Available at: https://www.datacamp.com/tutorial/r-web-scraping-rvest [Accessed: 30 October 2026].

Matanga, Y. 2020. *Big data analytics: practical handbook*. Faculty of Applied Science, Department of Information Technology. Unpublished course handbook.

Medium. 2025. How and Why to Add JSON-LD Structured Data to Your Website. *Medium*, 30 October 2025. [online] Available at: https://medium.com/@pavance40/how-and-why-to-add-json-ld-structured-data-to-your-website-f1aa19447388 [Accessed: 30 October 2026].

Oracle. n.d. *Class Long (Java Platform SE 8)*. Oracle. [online] Available at: https://docs.oracle.com/javase/8/docs/api/java/lang/Long.html [Accessed: 18 October 2026].

R CHARTS. 2026. Colors in R. *R Charts*, 2026. [online] Available at: https://r-charts.com/colors/ [Accessed: 15 Sep 2026].

R Core Team. n.d. *Pattern matching and replacement (grep)*. R Documentation. [online] Available at: https://search.r-project.org/R/refmans/base/html/grep.html [Accessed: 17 October 2026].

Robinson, L. 2023. Connect Database with R: Using PostgreSQL as database. *RPubs*, 17 April 2023. [online] Available at: https://rpubs.com/LoanRobinson/connect_database_R [Accessed: 6 November 2026].

Schweinberger, M. 2026. Automated Text Summarisation with R. *LADAL*, 2026. [online] Available at: https://ladal.edu.au/tutorials/text_summarisation/text_summarisation.html [Accessed: 31 October 2026].

Schweinberger, M. 2026. *Topic Modelling with R*. The Language Technology and Data Analysis Laboratory (LADAL), The University of Queensland. doi:10.5281/zenodo.19332978. [online] Available at: https://ladal.edu.au/tutorials/topic/topic.html [Accessed: 24 October 2026].

Silge, J. and Robinson, D. 2025. Text Mining with R: A Tidy Approach. Chapter 6, Topic modeling. *tidytextmining.com*, 7 April 2025. [online] Available at: https://www.tidytextmining.com/topicmodeling [Accessed: 24 October 2026].

Soetewey, A. and Zanotta, P. 2023. Web scraping in R. *Stats and R*, 16 January 2023. [online] Available at: https://statsandr.com/blog/web-scraping-in-r/ [Accessed: 30 October 2026].

Stack Overflow. 2013. Function to extract domain name from URL in R. *Stack Overflow*, October 2013. [online] Available at: https://stackoverflow.com/questions/19020749/function-to-extract-domain-name-from-url-in-r [Accessed: 30 October 2026].

Stack Overflow. 2013. R: add title to wordcloud graphics / png. *Stack Overflow*, March 2013. [online] Available at: https://stackoverflow.com/questions/15224913/r-add-title-to-wordcloud-graphics-png [Accessed: 1 November 2026].

Stack Overflow. 2014. Python - RegEx for splitting text into sentences (sentence-tokenizing). Stack Overflow, September 2014. [online] Available at:

https://stackoverflow.com/questions/25735644/python-regex-for-splitting-text-into-sentences-sentence-tokenizing [Accessed: 31 October 2026].

Stack Overflow. 2016. Connecting R to postgreSQL database. *Stack Overflow*, December 2016. [online] Available at: https://stackoverflow.com/questions/40642657/connecting-r-to-postgresql-database [Accessed: 6 November 2026].

Stack Overflow. 2016. *Text summarization in R language*. Stack Overflow, April 2016. [online] Available at: https://stackoverflow.com/questions/36064946/text-summarization-in-r-language [Accessed: 31 October 2026].

Sun Lab. 2022. *Hadoop Streaming*. CSE6250 Big Data for Health, Fall 2022. [online] Available at: https://sunlab.org/teaching/cse6250/fall2022/hadoop/hadoop-streaming.html [Accessed: 17 October 2026].

Traininghub.IO. 2023. Hadoop MapReduce architecture. *Medium*, 21 August 2023. [online] Available at: https://medium.com/@traininghub.io/hadoop-mapreduce-architecture-7e167e264595 [Accessed: 17 October 2026].

Usman, M. 2025. Topic Modelling (LDA). *RPubs*, 2 January 2025. [online] Available at: https://rstudio-pubs-static.s3.amazonaws.com/1260890_03b3707a9ab74f5fbfe6d83b6f8fa8a9.html [Accessed: 24 October 2026].

W3Schools. n.d. R Data Frames. *W3Schools*, n.d. [online] Available at: https://www.w3schools.com/r/r_data_frames.asp [Accessed: 5 November 2026].

W3Schools. n.d. SQL AVG() Function. *W3Schools*, n.d. [online] Available at: https://www.w3schools.com/sql/sql_avg.asp [Accessed: 6 November 2026].

W3Schools. n.d. SQL COUNT() Function. *W3Schools*, n.d. [online] Available at: https://www.w3schools.com/sql/sql_count.asp [Accessed: 6 November 2026].

W3Schools. n.d. SQL GROUP BY Statement. *W3Schools*, n.d. [online] Available at: https://www.w3schools.com/sql/sql_groupby.asp [Accessed: 6 November 2026].

W3Schools. n.d. SQL ORDER BY Keyword. *W3Schools*, n.d. [online] Available at: https://www.w3schools.com/sql/sql_orderby.asp [Accessed: 6 November 2026].

Wickham, H., Çetinkaya-Rundel, M. and Grolemund, G. n.d. Web scraping. R for Data Science (2e). *r4ds.hadley.nz*, n.d. [online] Available at: https://r4ds.hadley.nz/webscraping.html [Accessed: 30 October 2026].